Making Education More Accessible & Reliable

AI-Driven Facilitator-Based Teaching : **Project #133**

Moksh Manglani & May Thazin | Supervisor: Prof Gill Dobbie

THE UNIVERSITY OF AUCKLAND
Te Whare Wānanga o Tāmaki Makaurau
NEW ZEALAND

Background

44 Million
More teachers needed worldwide by 2030 [1]

70%
of 10-year-olds in low and middle income countries cannot read a simple text [2]

Under 15%
finish unsupervised online courses [3]

	Personalisation	Decoupled delivery	Structured guidance	Accessibility	Adaptability	Engagement
Formal schooling	~	✗	✓	~	~	✓
Homeschooling, self-directed	✓	~	✗	✓	✓	~
MOOCs, online self-paced	~	✓	~	✓	~	✗
Blended, synchronous	~	~	✓	~	~	✓
LMS platforms	~	~	~	✓	✓	~
Community-based models	~	~	✗	~	~	✓
Our StudyBuddy	~	✓	✓	✓	✓	~

Figure 1. How existing education models compare on six criteria, based on our literature review

✓ strong · ~ partial · ✗ weak

Note: Decoupled Delivery depicts whether a lesson can be delivered without a teacher present

Two sides: admin-facilitator (designing and delivering the lesson) and facilitator-student (seeing who needs help).

Designing and Delivering the Lesson

(admin-facilitator side)

RQ: Can any approved facilitator, regardless of subject expertise, deliver an expert-designed lesson?

Methods

1	**Researched**	The problem, and what already exists.
2	**Understood**	What a teacher does, and their biggest problems.
3	**Designed**	Separating lesson **creation** and **delivery**
4	**Built**	A live platform with NZ and Indian curriculum lessons.
5	**Refined**	The Platform based on informal feedback.

Using these steps we made a platform that allows experts to create lessons and facilitators to deliver them without subject expertise whilst also supporting them.

Admin
Designs lessons

Facilitator
Delivers them

↔

Web app
Next.js on Vercel
Checks every action on the server
Steps unlock in order
AI drafts worksheets; admin approves

↔

Supabase
Login · File storage
Database
Organisation → Class → Subject → Lesson → Block
Facilitator · Lesson progress

To test our design we got feedback through informal chats. We also used various HCI frameworks to evaluate our UI/UX and System design, like the Nielsen's Heuristics. We used these as a guide to improve usability.

Nielsen's 10 usability heuristics
Reviewed the whole platform against all ten. Three examples:

Visibility of system status	Each lesson step shows done, active or locked, plus live progress
Match the real world	Class cards look like Canvas courses, a layout users already know
User control and freedom	Deleting a lesson block needs a second click to confirm

Results

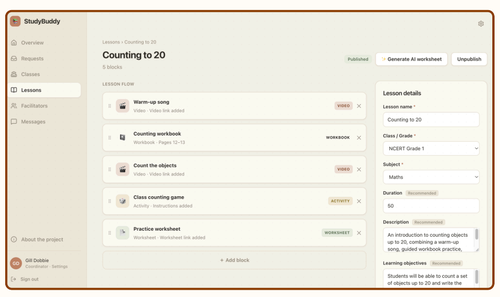
Admin
creates the lesson using blocks like- videos, PDFs, workbooks and resources

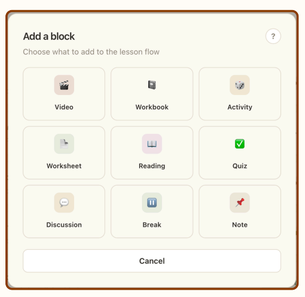
→

In Classroom
the lesson is played by the facilitator on a big digital screen, the platform teaches the students

→

Facilitator
navigates the lesson, answers questions, and ensures everything goes smoothly





Try It Yourself!

Helping Facilitators Support Students

(facilitator-student side)

RQ: Can a software system convert learner interaction data into evidence-linked recommendations for facilitator decision-making?

Methods

- Our system turns a short reading activity into evidence-linked flags, showing facilitators which students need support and why. We focused on reading comprehension due to the "4th-grade slump" (~Yr 5 NZ), when many children start falling behind. [4]
- Reading needs two skills: decoding words and understanding them. [5] The activity asks detail and inference questions, then "Did you get it right?"

Student Learner Activity → **Records Student Data** → **Rule Engine Classification** → **Facilitator Insights**

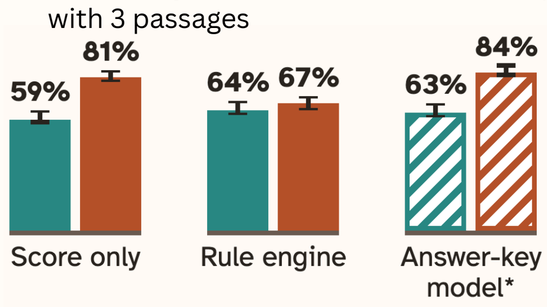
Five simulated reader types (assumed % right: ● detail ● inference)

90 · 80	**Typical:** usually right on both question types
90 · 82	**Careful reviser:** rereads and fixes answers
85 · 45	**Poor comprehender:** (about 1 in 9 children) [6]: right on detail, wrong on inference, often thinks they are right
55 · 45	**Decoding-limited:** slow, asks for hints before trying, knows they are struggling
50 · 35	**Weak in both:** low on both question types

Tested on 200 simulated classrooms of 30, on separate simulated learners. Each learner read 1 or 3 short passages (4 questions each). We compared our rule engine with score only (flag anyone below 3 of 4 right) and an answer key model*, 95% intervals.

Results

- **One passage**: our rule engine catches 64% (about 4 of 7 in a class of 30), which is more than the score only model (59%) and level with the answer key model* (63%)
- **Three passages**: score only and the answer key model* catch more (81%, 84%) than our rule engine model (67%), but its flags are correct 94% of the time (score only flags 72% correctly)
- **Poor comprehenders**, the hardest to spot, drop from 54% with 1 passage to 39% with 3 passages



***Answer-key model**: a model given the true answers, which no real classroom has. Most accurate overall (about 2% false alarms), so it flags fewer borderline learners. A reference, not a usable tool.

Findings & Conclusion

Separating who designs a lesson from who delivers it lets any approved facilitator deliver it as designed, and our rule engine shows them whom to help.

From school practitioners: teaching quality varies between classes, and continuity is lost when a teacher leaves; managing a classroom matters as much as knowing the subject

64%
of learners who need support are **caught from one short task** (~4 of 7).

94%
of our flags are **correct with three passages** (increased from 67% with one passage)

Limitations

- StudyBuddy has not been tested in a real classroom yet.
- Lesson quality depends on the admin's lesson creation skills and resources available to the facilitator like devices and reliable internet.
- Simulated learners are based on research & broad assumptions.
- Rule engine was set manually and adjusted once after first run, doesn't yet combine passages effectively (variability limited).

Future Research

- Test StudyBuddy in a real classroom and measure its impact on learning.
- Explore more ways to guide and support the facilitator in delivering higher-quality lessons.
- Collect data from varied passages, tune how the rule engine combines them, and check its flags against teachers' judgements in a real classroom pilot.
- Modular design: the same process could be adapted to other subjects.

References

[1] UNESCO (2024). Global report on teachers: Addressing teacher shortages and transforming the profession.
[2] World Bank (2022). The state of global learning poverty: 2022 update.
[3] Jordan, K. (2015). Massive open online course completion rates revisited: Assessment, length and attrition. IRRODL, 16(3), 341–358.
[4] Chall, J. S., & Jacobs, V. A. (2003). The classic study on poor children's fourth-grade slump. American Educator, 27(1), 14–15, 44.
[5] Gough, P. B., & Tunmer, W. E. (1986). Decoding, reading, and reading disability. Remedial and Special Education, 7(1), 6–10.
[6] Kelso, K., Whitworth, A., Parsons, R., & Leitão, S. (2022). Hidden reading difficulties: Identifying children who are poor comprehenders. Learning Disability Quarterly, 45(3), 225–236.